International Review of Financial Analysis 96 (2024) 103721

Contents lists available at ScienceDirect

International Review of Financial Analysis

journal homepage: www.elsevier.com/locate/irfa

Social media attention and retail investor behavior: Evidence from r/wallstreetbets☆

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ARTICLE INFO

JEL classification:
G11
G41

Keywords:
Risk-taking
Investor welfare
Social media
Reddit
Emotions
Natural-language-processing

ABSTRACT

This paper studies the influence of Reddit social media attention on investor risk-taking and welfare using individual trading data from a large trading platform. We show that attention generated on the subreddit r/wallstreetbets (WSB) spurs uninformed trading, also in short positions. WSB attention increases overall risk levels as investors trade riskier stocks and allocate larger portfolio shares to new positions. WSB attention significantly reduces holding period returns (HPRs). Positions created when WSB attention is at its highest realize −8.5% HPRs, while the average HPR across all investments is positive. Investment decisions of retail investors are more likely to be based on emotional responses to angry and disgusted user contributions on Reddit rather than analyst investment advice.

1. Introduction

Individual investors make financial decisions that are influenced by behavioral biases. They trade too much, underdiversify, herd, sell winners too soon while holding on to losers too long (see, *e.g.*, Nofsinger & Sias, 1999; Odean, 1999; Weber & Camerer, 1998), and suffer from limited attention (Barber, Huang, Odean, & Schwarz, 2022; Glaze, 2022; Hsieh, Chan, & Wang, 2020). Time and cognitive constraints hinder individual investors from processing all information on available investment options, so high-attention investments are salient and more likely to be considered (Barber & Odean, 2008). As a result, individual investors are net buyers of attention-grabbing stocks, which is considered an investment mistake (Mavruk, 2017; Seasholes & Wu, 2007). Constrained by limited attention, individual investors increasingly rely on social media platforms to gather investment advice from their peers. With increasing concerns from regulators, more and more (young) investors rely on social media to obtain investment advice. In 2023, almost 80% of american millenial and Gen Z investors relied on investment advice from social media outlets.[1]

The subreddit WallstreetBets (WSB) is a prominent example of a social media outlet that investors use to obtain investment advice. In early 2021, the subreddit made headlines when it was associated with an investment mania that stirred up conventional finance.[2] Apparently, the episode was caused by the 2020 increase in accounts at low-fee brokers, which were subsequently encouraged by alleged financial experts on WSB to bet onto stocks such as GameStop (GME). The WSB platform describes itself as "[...] a community for making money and being amused while doing it. Or, realistically, a place to come and upvote memes when your portfolio is down".[3] The platform has gained popularity through its humorous memes, which also served as the mania's namesake: *meme-stock mania*. As a result, stocks with an increase in trading activity that is not justified by a change in fundamentals but rather because of sudden attention on social media have since been called meme stocks.

When discussing meme-stock mania, the media and empirical literature have mainly painted a picture of manipulative millennial investors (Corbet, Hou, Hu, & Oxley, 2022) who engage in high-risk speculative trading (Hasso, Müller, Pelster, & Warkulat, 2022), distort asset prices, and impede corporate funding. Contrarily, abnormal

☆ We are grateful for very helpful comments and suggestions from Samuel Vigne (the editor), three anonymous reviewers, and participants of the Brown Bag Seminar at Paderborn University and the HVB doctoral seminar 2022. Data were obtained under a non-disclosure agreement with a financial institution. We thank our data provider for the use and explanations of their data. Any errors, misrepresentations, and omissions are our own.

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[1] https://www.forbes.com/advisor/investing/financial-advisor/adults-financial-advice-social-media/

[2] https://www.economist.com/finance-and-economics/2021/01/30/the-frenzied-rise-of-gamestop or here https://www.economist.com/the-world-ahead/2021/11/08/the-phenomenon-of-meme-stocks-could-be-here-to-stay.

[3] https://www.reddit.com/r/wallstreetbets/wiki/faq/

https://doi.org/10.1016/j.irfa.2024.103721
Received 21 May 2024; Received in revised form 14 August 2024; Accepted 27 October 2024
Available online 4 November 2024
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stock returns of companies that were the target of WSB pump and dump schemes are positive (Corbet et al., 2022). Companies such as GameStop heavily capitalized on the frenzy with stock sales[4] or drew on lines of credit.[5] Brokerage services most likely profited from the increase in trading volume, either through trading costs or payment for order flow.

This paper focuses on WSB social media attention and individual retail investors. We analyze whether attention on the WSB subreddit increases investor risk-taking and affects investor welfare. While Cookson, Lu, Mullins, and Niessner (2024) find that attention from Twitter, StockTwits, and Seeking Alpha is highly correlated across platforms, the literature also provides suggestive evidence that attention is indeed different on WSB (Cahill, Liu, & Smales, 2022). Investors at low-fee brokerages have a reputation of being easily influenced by social media attention that originates on the subreddit and consequently make irrational financial decisions that harm their financial welfare and contribute to investment manias (Randall, 2021). Thus, the unique features of the WSB subreddit require in-depth analyses. Consequently, this paper asks whether WSB attention influences investor risk-taking and welfare.

We quantify WSB attention with the frequency of abnormal ticker or company mentions on the WSB subreddit. Then, we exploit individual trading data from an online brokerage platform and relate these mentions to trading characteristics of retail investors. The broker allows its investors to take levered positions, short positions, and, on average, charges low transaction costs. Account-level data allow us to analyze trading characteristics and the welfare of the new subclass of retail investors in detail. Research on the effects of attention on individual investor behavior and individual retail investor welfare in the context of meme-culture remains limited. Importantly, we do not have to rely on algorithms that potentially misclassify a number of trades (Barber, Huang, Jorion, Odean, & Schwarz, 2024) to identify retail trades. Furthermore, the data allow us to observe detailed trading patterns such as holding periods and net profits without relying on proxy variables. We study all meme stocks – *i.e.*, all stocks that have experienced sudden attention on social media – that are tradable in the broker's universe.

We first briefly look at the relationship between WSB attention and net buying of both long and *short* investors. In line with the existing literature (Avila, Martineau, & Mondria, 2023; Bradley, Hanousek, Jame, & Xiao, 2024; Cookson et al., 2024; Hu, Jones, Zhang, & Zhang, 2023), we find that WSB attention predicts buy–sell order imbalances for our sample of retail investors. Adding to the existing literature, the individual trade data allow us to show that WSB attention also increases short-selling by retail investors. Next, we dig deeper and study the direct relationship between attention and trade characteristics such as investor risk-taking. We observe that investors who trade on WSB attention trade riskier stocks, but compensate the increased stock risk by taking lower leverage. Further, we observe WSB attention trading to reduce investors' trading profits. Finally, we study whether investment decisions tend to be driven by emotional responses to user contributions on the subreddit or by analyst investment advice. Our results indicate that mostly angry and disgustful posts impact trading patterns.

Emotions have important implications for financial decision-making (Long, Lucey, & Yarovaya, 2023; Taffler, 2018; Taffler & Tuckett, 2010; Vamossy, 2021). Taffler (2018) argues that "[i]nvestment decisions are driven jointly by emotion and cognition with most associated mental activity unconscious" (p. 649). Emotions also play an important role on social media. Importantly, emotions expressed in social media posts may impact the emotional states of others (Ferrara & Yang, 2015). While the literature on emotional finance agrees that emotions impact financial markets, the literature does not agree whether positive or negative emotions have a greater influence. Our results support those studies that observe an effect of negative emotions in particular. For example, Nguyen, Calantone, and Krishnan (2020) find that transient investors decrease their stockholdings in response to increases in negative emotions such as anger, disgust, and fear.

This paper makes several contributions to the literature. First, we add to the literature on the informativeness of retail trading patterns in the context of social media attention (see, *e.g.*, Bartov, Faurel, & Mohanram, 2018; Bradley et al., 2024; Chen, De, Hu, & Hwang, 2014; Cookson, Engelberg, & Mullins, 2023). We contribute to this literature by exploiting individual trading data from a large online brokerage platform to quantify retail trading. The individual trading data allow us to study trade characteristics and outcomes at the micro level. We show that retail-trading patterns explained by WSB attention constitute poor trading decisions. In particular, investors who buy stocks when WSB attention is at its highest suffer negative trading outcomes. Second, the paper contributes to the literature that analyzes attention and retail investor trading characteristics (see, *e.g.*, Arnold, Pelster, & Subrahmanyam, 2022; Davydov & Peltomäki, 2023; Gargano & Rossi, 2018). Using WSB to proxy social media attention, we are able to add a new attention dimension to this literature. We find that WSB attention trade are associated with higher stock volatility and larger portfolio weights, and negative profits. We infer that meme attention promotes speculative behavior that results in negative profits for meme investors. Finally, we contribute to the literature that analyzes the impact of social media sentiment on trading behavior (see, *e.g.*, Bollen, Mao, & Zeng, 2011; Cookson et al., 2024; Duz Tan & Tas, 2021; Farrell, Green, Jame, & Markov, 2022; Gu & Kurov, 2020; Hu et al., 2023). We add to this literature by not only focusing on sentiment, but rather take the emotions of social media posts into account. Our findings indicate that investment decisions of meme investors are more likely to be based on emotional responses to angry and disgusted user contributions on social media than analyst investment advice. We discuss the various streams of the literature in more detail in Section 3.

The remainder of this paper proceeds as follows. First, Section 2 elaborates on meme-stock mania and its implications for the investment landscape. Then, Section 3 discusses the related literature. Section 4 develops the hypotheses. Section 5 describes the data and introduces the variables for the empirical analyses. Section 6 presents the main analysis and discusses the results. The final section concludes the paper.

2. Meme-stock mania

Meme-stock mania made headlines in early 2021, but interest persists. For example, Roundhillinvestments built the $Meme, a Meme Stock ETF that tracks the performance of meme stocks, and Bloomberg tracks and regularly reports on the performance of a basket of 37 meme stocks.[6] In their 2022 outlook, the economist argues that the meme-stock phenomenon could be here to stay and even intensify,[7] which was corroborated by Bloomberg journalists in 2022.[8]

In early 2021, when markets supposedly had been driven into meme market territory by social media, regulators articulated concerns about market efficiency and retail investor welfare (Bradley et al., 2024). For example, Massachusetts' top securities regulator, William Gavin, proposed a 30-day trading halt to protect "small-time" and "unsophisticated investors" from reckless risk-taking that he regarded as driven

[4] https://www.cnbc.com/2021/06/22/gamestop-jumps-9percent-after-the-original-meme-stock-cashes-in-again-with-1-billion-share-sale.html

[5] https://www.bloomberg.com/news/articles/2022-08-28/meme-stock-mania-is-back-and-wall-street-says-it-s-here-to-stay

[6] https://www.bloomberg.com/news/articles/2022-03-29/meme-stock-mania-returns-with-big-swings-and-retail-money-flood

[7] https://www.economist.com/the-world-ahead/2021/11/08/the-phenomenon-of-meme-stocks-could-be-here-to-stay

[8] https://www.bloomberg.com/news/articles/2022-08-28/meme-stock-mania-is-back-and-wall-street-says-it-s-here-to-stay

by the fear of missing out (FOMO) and investor incompetence.[9] Furthermore, popular online brokerages such as TD Ameritrade, Charles Schwab, Interactive Brokers, or Robinhood temporarily restricted trading on certain meme stocks with the argument of protecting the market. Unsurprisingly, this encountered strong headwinds as retail investors on online platforms kept insisting to be "every bit as sophisticated as institutional investors",[10] and felt to be missing out on investment opportunities.

Why do retail investors turn to social media for investment advice in the first place? The U.S. Securities and Exchange Commission (SEC) statutory mandate is to maintain fair, orderly, and efficient markets. Up until recently, however, large investors possessed an informational advantage over individual investors, which was mainly due to the costs of information procurement (Chiu & Yahya, 2022; Engelberg, Reed, & Ringgenberg, 2012; Larcker & Lys, 1987; Massoud, Nandy, Saunders, & Song, 2011). With the internet as an information intermediary, that information asymmetry seems to have started to dissolve (Drake, Thornock, & Twedt, 2017; Grennan & Michaely, 2021). In recent years, retail investors have started to build their own easy-to-acquire sources of information on online platforms and in chatrooms (Chiu & Yahya, 2022). This includes semiprofessional platforms such as Seeking Alpha or Estimize that crowdsource investment advice from professionals, practitioners, and individuals and offer the advice at a reasonable rate.[11] Lower-cost information sources also include social media such as Facebook, Twitter/X, Youtube, and Reddit (*e.g.*, Bartov et al., 2018; Buz & de Melo, 2021; Hasan & Cready, 2019; Heimer, Simon, et al., 2012; Xiao, 2020).[12] Despite these new sources of information, disparities continue to exist. Institutional investors monitor stocks with the help of portfolio risk managers and financial analysts' reports. In contrast, individual investors have the reputation of feeling the *mood* of the market by engaging in discussions on social media platforms (Costola, Iacopini, & Santagiustina, 2021).

Not only has technology improved the access to and supply of financial information, it also has lowered barriers to enter financial markets for individual investors. The main factor that led to lower those barriers is the digitization and decentralization of asset management, from human advisers to gamified trading apps (Chapkovski, Khapko, & Zoican, 2024). As a consequence, retail trading volume surged. In 2021, U.S. retail trades constituted 22% of total stock trades,[13] and in 2023 neo-brokerage accounts were held by more than 20% of the US adult population (ESMA, 2024). The increase in demand is among other things attributed to the easy-to-use apps and the promise of low-commission fees of neo-brokerages (see e.g. Aramian & Comerton-Forde, 2023). Thus, the access to retail trading has never been easier and retail trading has never looked more attractive.

The ease of trading securities, low-cost access to financial information, and cost-effective trading conditions paired with a new mindset are a game-changer in individual investor behavior. Additionally, as retail trading volume represents up to one-fifth of US equity volume, it is of utmost importance to more closely examine retail investor welfare in the context of social media for both regulators and investors.

3. Related literature

As briefly eluded to in the introduction, we contribute to several strands of the literature. First, our paper is related to the literature that analyzes the effects of information supply on retail investment value. On the one hand, this literature finds that financial media and crowd-sourced investment platforms provide investment value (Bartov et al., 2018; Chen et al., 2014; Jame, Johnston, Markov, & Wolfe, 2016). On the other hand, research indicates that investor social networks enhance behavioral biases and spread stale news (Cookson et al., 2023). In this context, it is important to consider that not all crowd-sourced investment platforms are alike. Several platforms such as Seeking Alpha or Estimize take charges, while others are free. Perhaps as a result of varying charges, different platforms also attract a different audience. Thus, it is important to carefully investigate novel platforms. In line with this notion, a novel literature focuses on Reddit. Closely related to our paper, Bradley et al. (2024) show that informativeness in due diligence reports vanished following the GME short-squeeze. We add a detailed analysis on trade characteristics and the profitability dimension to this analysis. In a contemporaneous paper, Eliner and Kobilov (2023) find that returns to social media-induced trades, *i.e.*, those involving a stock that has abnormally high discussion activities on WSB relative to a baseline period, are lower than returns on all other positions opened on the same day.

Second, the paper is related to the literature that links attention and individual investor behavior. This literature has brought forth various proxies for attention. Arnold et al. (2022) distinguish between endogenous and exogenous attention. In line with previous studies from experimental psychology and neuroscience, they argue that *endogenous attention* refers to the process to deliberately deal with a matter, whereas *exogenous attention* refers to the process by which external stimuli involuntarily redirect an individual's attention. Perhaps the most widespread proxy for endogenous attention is abnormal Google search volume (ASVI) introduced by Da, Engelberg, and Gao (2011). ASVI and similar proxies capture individuals efforts to obtain information on a particular topic. Several studies exploit investors' logins to their brokerage accounts as an endogenous attention measure to capture investors' willingness to deal with their investment decisions and monitor their portfolios (Davydov & Peltomäki, 2023; Gargano & Rossi, 2018). In this literature, Gargano and Rossi (2018) measure overall portfolio attention using investors' account logins and find that endogenous attention is positively related to investment performance while Dierick, Heyman, Inghelbrecht, and Stieperaere (2019) find that attentive investors trade less in line with the disposition effect. Davydov and Peltomäki (2023) find that investors who exploit different sources of investment leverage underperform investors who do not; however, they show that this negative effect of leverage usage decreases with greater investor attention. Arnold et al. (2022) use push notifications as exogenous measure of attention and find that those (exogenous) attention triggers increase investor risk-taking.

Social media attention in general and WSB attention in particular are somewhere between endogenous and exogenous measures of attention. While some investors may actively visit social media pages to obtain novel information and to deliberately study investment opportunities (*i.e.*, endogenous attention), others may receive push notifications from their social media apps and have they attention redirected toward a particular stock (*i.e.*, exogenous attention). Even when investors visit social media pages to obtain information, their attention may be redirected as the most recent discussions may focus on different stocks than those that the investors wanted to research. Thus, we consider WSB attention to be closer to exogenous attention than to endogenous attention.

With respect to social media attention, Cookson et al. (2024) and Hu et al. (2023) show that social media attention predicts next-day retail order imbalance while Avila et al. (2023) and Bradley et al. (2024) show that WSB predicts same-day retail order imbalance. While these

[9] https://www.cnbc.com/2021/01/27/gamestop-speculation-is-danger-to-whole-market-massachusetts-regulator.html

[10] Reddit user *sleeksleep* on WSB on 2021-01-30.

[11] Seeking Alpha starts at USD 240 p.a., while Estimize charges USD 325 for 3 months (20/06/22).

[12] A third group that offers investment strategies is a collection of Telegram and Discord groups. However, they primarily focus on unregulated cryptocurrency markets and mainly promote pump-and-dump schemes (Hamrick et al., 2021; Li, Shin, & Wang, 2021).

[13] https://www.forbes.com/sites/dereksaul/2023/02/03/retail-trading-just-hit-an-all-time-high-heres-what-stocks-are-the-most-popular/?sh=1d01c5316664

papers make use of proxies, we are able to explicitly quantify retail trading using individual trade data from a large online brokerage platform and thereby focus on trade characteristics and the profitability of WSB attention trades.

Thereby, this paper also adds to the literature that studies the profitability of retail trades (see also Eliner & Kobilov, 2023). Until recently, researchers drew conflicting conclusions. For example, Odean (1999), Grinblatt and Keloharju (2000), and Barber and Odean (2008) associate retail trades with weak investment performance, whereas (Kaniel, Saar, & Titman, 2008), Kaniel, Liu, Saar, and Titman (2012), and Boehmer, Jones, Zhang, and Zhang (2021) find a positive correlation between retail order imbalance and returns. Barber, Lin, and Odean (2021) find that retail trades positively predict returns but are not profitable. They explain this paradox with the argument that stocks that have experienced high demand from retail investors tend to underperform subsequently.

Finally, there is also a growing strand of literature that has started to analyze the meme domain. Long et al. (2023), for example, study the impact of sentiment on GameStop intraday returns and find fear to be the dominant sentiment. Costola et al. (2021) find common stylized facts of meme stocks and provide a general methodology to identify *mementum*, the meme periods of stocks. Bradley et al. (2024) and Buz and de Melo (2021) investigate the quality of investment advice that is shared on WSB and how it affects investors. Focusing on sentiment, Farrell et al. (2022), Hu et al. (2023), and Cookson et al. (2024) study the relationship between sentiment and same-day and next-day order imbalance. Anand and Pathak (2022) focus on GameStop and provide evidence that the tone of the discussions on WSB has an impact on stock market characteristics. Suchanek (2024) extends this evidence to other meme stocks. Fedyk (2022) relies on the Robintrack dataset that ends in mid-2020 and shows that the daily number of RobinHood investments is predicted by WSB posts. Corbet et al. (2022) argue that the "will of sarcastic Millennials and memes" could disturb the fairness of securities transactions, deteriorate market companies, and propagate financial bubbles. The authors identify pump and dump events in WSB postings and show that manipulative investor attention has a positive effect on abnormal returns and liquidity. Exploiting retail trading data from a large online brokerage platform allows us to contribute to this strand of the literature and connect social media activity with retail trade characteristics and profitability.

4. Hypotheses

We begin by discussing the relationship between attention and retail trading characteristics. Attention has been shown to influence trading outcomes in two opposite directions. Rational models suggest that investors should profit from paying (endogenous) attention (see, *e.g.*, Kacperczyk, Van Nieuwerburgh, & Veldkamp, 2016; Van Nieuwerburgh & Veldkamp, 2010). In accordance with these findings, Gargano and Rossi (2018) empirically show that paying attention to one's own portfolio increases profitability (see also Davydov & Peltomäki, 2023, for similar results). At the same time, exogenous attention triggers such as push notifications in trading apps have been shown to increase risk-taking and trading intensities among retail investors (Arnold et al., 2022). Arnold et al. (2022) find no effect of exogenous attention on investors' individual trade performance.

While these results may seem contradictory, they can be explained by dual-process theories (*e.g.*, Alter, Oppenheimer, Epley, & Eyre, 2007). Humans use two distinct systems to process information: System 1 (fast) and System 2 (slow) (Kahneman, 2011). System 1 is intuitive, fast, and automatic and described as non-statistical and heuristic. System 2, on the other hand, requires effort. It is conscious, slow, controlled, statistical, and cautious. While System 1 may generate fantastic ideas, it also may result in systematic errors (Shleifer, 2012).

When making investment decisions, ideally people carefully weigh investment options and spend some time and effort on it. Some might even employ statistical analyses to maximize their payoffs. Thus, it is reasonable that an investors' endogenous search for financial information operates under System 2. It is controlled, slow, effortful, and maybe in some cases even statistical. Since System 2 thinking is less likely to result in systematic biases, the endogenous attention toward stocks is, on average, beneficial. However, if that endogenous search is interrupted by external factors, attention is redirected. Investors no longer reason with System 2 but fall back to fast, intuitive, and heuristic decision making that may lead to rash financial decision-making with suboptimal outcomes.

We argue that attention generated on WSB is an exogenous attention trigger. While Reddit allows users to search for keywords within specific channels and throughout the platform, the default website settings display the *hot* posts on the start page. Hot posts refer to comments that are new and are becoming popular fast.[14] Additionally, users can join a subreddit and as a result receive desktop notifications and updates. Thus, users do not necessarily have to actively search for information, because that information is sprung upon them, either through the website's design or notifications. Moreover, investor activities on WSB have coverage across platforms (Costola et al., 2021), implying that WSB attention may even extend to the customized (news) feeds on people's phones or laptops.

Additionally, the word *information* has to be used carefully in this context. While some user contributions, for example, the due diligence reports used by Drake et al. (2017), might provide valuable information, a large part of the content shared on WSB does not qualify as new information.[15] Thus, we hypothesize:

H1: WSB attention increases investor risk-taking.

We base our next hypothesis on noise trader theory (see, *e.g.*, Abraham, Elan, & Marcus, 1993; Brown, 1999; Lee, Shleifer, & Thaler, 1991). The theory's main implication is that irrational investors who trade collectively on a noisy signal are able to induce risk in the market. Noise trading thus may cause prices to drift away from fundamental values. Arbitrageurs may choose not to correct prices because of the possibility that the noise traders' asset demand will increase and cause losses for anyone trading against them (De Long, Shleifer, Summers, & Waldmann, 1991). Thus, prices will continue to deviate from their fundamental values. The additional risk, stemming from irrational noise traders that act in concert, should be priced in the market and increase volatility (Brown, 1999).

Building on noise trader theory, Boehmer et al. (2021) raise the question of whether retail equity investors are informed, *i.e.*, whether they can predict future stock returns. The authors find that marketable retail order flow has predictive power for future returns. They provide suggestive evidence that the retail order flow contains relevant information that is not yet incorporated into stock prices. These findings are supported by, for example, Kaniel et al. (2012), Kelley and Tetlock (2013), and Barrot, Kaniel, and Sraer (2016). Conversely, others find that retail investors are uninformed and make systematic mistakes in their investment decisions (Barber et al., 2022; Barber & Odean, 2000, 2008; Foucault, Sraer, & Thesmar, 2011).

Due to recent developments in financial market accessibility and the increasing popularity of trading apps, the investment landscape has changed significantly. News coverage suggests a new level of irrational behavior in retail investors that is already supported by initial empirical evidence (Barber et al., 2022). Although the evidence on whether retail investors make informed trading decisions, *i.e.*, whether their trading patterns can predict future stock returns, remains diverse, but not conclusive, we argue that recent developments have favored the strand of literature that argues in favor of uninformed retail investors. Moreover, we argue that uninformed trading is spurred by attention,

[14] Other available sortings include new, top (comments with the most upvotes and replies), and rising.

[15] We provide a selection of posts in Table 3.

that is, investors make systematic mistakes when they collectively trade on the noisy signal attention. Thus, we hypothesize the following:

H2: Attention on WSB spurs uninformed trading and harms investor welfare.

Finally, we study whether meme investors tend to make emotional rather than informed decisions when managing their investments. Analyst recommendations have been found to add value for investors. For example, Womack (1996) finds that stocks with the most affirmative recommendations outperform those with the least affirmative recommendations (see also Jegadeesh & Kim, 2006). Conversely, investor emotions have been shown to contribute to the development of financial bubbles (Andrade, Odean, & Lin, 2016; Breaban & Noussair, 2018) and to result in smaller earnings announcement returns (Vamossy, 2021). The access to professional investment advice is limited for retail investors, mainly due to financial reasons. Alternatives have emerged, that to a large degree, consist of social media platforms. These platforms are interactive and allow users to not only passively assimilate information but also to actively participate in discussions.

We argue that meme investors' investment decisions are more likely to be based on emotional responses to comments on social media than on analyst investment advice. Thus, we hypothesize the following:

H3: Emotional sharing on WSB is a predictor of investor trading patterns.

H4: Analyst investment advice cannot predict investor trading patterns.

5. Data and variables

5.1. Data

We use transaction-level data from a large online brokerage service. Data were obtained via a non-disclosure agreement. The brokerage service, which classifies as a neo broker, allows its customers to trade various stocks, ETFs, foreign exchange, and even cryptocurrencies. Investors can trade via the brokerage website and an app. The average investor is male and between 25 and 34 years old. The brokerage data contain the entirety of trades carried out between January 1, 2020, and February 12, 2021. In total, the sample includes over 100 million trades executed by over 1.4 million investors, covering 2,647 stocks. We observe the timestamp of each trade (opening and closing of positions), whether the trade was a long or short position, leverage, the net profit, and investments (as a percentage of portfolio value). Each investor comes with a unique identifier, making all trades traceable. The data also contain investor-specific characteristics such as gender and age.

We collect equity data from Refinitiv. We obtain prices, trading volume, market capitalization, price-to-book ratios, ticker, and company names from Eikon and match them to the trading data based on their ISIN. We collect analyst recommendations and earnings per share estimates from I/B/E/S and link the data via ticker symbols. We complement the data with company-related event data from RavenPack. We collect the daily event volume for each company as a proxy for the degree of media attention a company receives and additionally download the Entity Sentiment Indicator (ESI) to account for whether the event is perceived as positive or negative (Hafez, Lautizi, Guerrero-Colon, Gomez, & Matas, 2018). We exclude financial companies from the analyses.

We download user contributions on the WSB subreddit via pushshift, a social media data collection platform that makes Reddit data available to researchers.[16] The sample contains all available posts (titles and comments) from the WSB subreddit. In total, they represent more than several million user contributions. Because investor activities have coverage across platforms (Costola et al., 2021), the exposure group of WSB user contributions is not limited to its current 12.5 million followers but extends to everyone with access to the internet.

[16] https://github.com/pushshift/api

5.2. Variables

The main variable of interest is *attention*. We measure attention with ticker and/or company name mentions. As a starting point to identify ticker mentions, we scan all posts for cashtags ("$"). Cashtags have their roots in Twitter/X and operate like hashtags #. Users can click on cashtagged stock symbols and obtain the twitterverse's opinion. The habit of tagging stocks has spilled over to various other social media platforms such as WSB. We search all posts for $ signs and save the five characters following. Since ticker lengths differ across exchanges, we use the longest. NASDAQ securities have ticker symbols of up to five characters.

After excluding all noncharacter elements from the cashtag searches (*i.e.*, $ often comes with a number behind it), we match these potential stock tickers to the tickers included in our trading data. This results in a total of 1,359 stocks that are used for the analyses. Those tickers are matched with the company names from Eikon. Then, we extract all WSB user contributions that contain any of the 1,359 stock tickers and company names previously identified. We carefully search the data for various versions of the stock ticker. Stock symbols with character lengths greater than two are handled as follows. The versions for GameStop would include: $gme, $g m e, $g-m-e, $g.m.e, gm$e, gme, gamestop, g m e, g.m.e, and g-m-e. We transform all WSB posts to lower case and use lower case letters for the search only. For symbols with length smaller than three, the versions gme and g m e would be excluded to avoid capturing abbreviations. Additionally, we manually check all ticker and company names to ensure that they have no meaning other than a company context.[17] In cases where this could be an issue, such as CAT, we limit the searches to cash-tag-associated searches only. If company names are affected, company names are excluded from the search, or where appropriate, an addendum is added such as "holding".[18] Finally, we exclude posts that we deem to distort the results such as posts that include more than 30 ticker symbols, posts that have patterns such as $a $b $c ..., and administrative posts that are identifiable by author.

Since we are interested in how the sudden increase in attention on Reddit affects retail investors, we use abnormal attention as our main variable of interest. Abnormal attention is defined as the number of WSB posts for a specific stock on a certain date minus its 30-day moving average. Note that posts can be published and read 24/7, as opposed to equities trading, which is limited to trading hours. Thus, when creating lag-structures, we use the average of Friday, Saturday, and Sunday as Monday's lag.

To benchmark WSB attention against other attention measures, we additionally consider Abnormal Google Search Volume (ASVI, see Da et al., 2011) as a proxy for endogenous (market-wide) attention and the common component of attention from Twitter, StockTwits, and Seeking Alpha following Cookson et al. (2024) as an alternative measure for social media attention (*social attention signal*). Cookson et al. (2024) find that attention from Twitter, StockTwits, and Seeking Alpha is highly correlated across platforms.

To answer the research questions, we furthermore utilize the assets' *returns*, market *volume*, and *volatility*. Returns are defined as the daily assets' log-returns, and volume is the daily number of traded shares. Volatility is estimated with an ARMA(p,q)+GARCH(1,1) model with automatic parameter selection and student-t innovations.

We further include the retail *buy–sell imbalance (BSI)* of executed trades defined as the relation between long minus short to total positions. Note that there are two possibilities to take long and short positions on a stock. For long positions, investors can either open a long

[17] This does not mean that we read all user contributions. Instead we checked a randomly drawn sample of 20 contributions per ticker.

[18] The emotion recognition samples in Table 2 highlight the importance of including the company stock ticker without associated cashtags.

position or close a short sale. Short positions can be taken by closing a long position or selling a stock short. In addition to analyzing the aggregated BSI of long and short trades, we separately compute the BSI of long (*BSI long*) and short trades (*BSI short*). BSI long is defined as the relation between long minus short to total positions, but that long position can now only be taken by opening a long position and not by closing a short sale. The same applies for BSI short. We count all trades in a given stock that we observe in our dataset to estimate BSI. We separately study long and short positions based on the notion that attention may have a different effect on the buying and selling behavior of retail investors. Separately studying BSI allows us to shed light on this notion.

As a counterpart to market volume, we also examine *trading intensity*, which is the number of retail trades for a specific asset on a given day. Again, we count all trades in a stock on a given day. Investor-specific variables are investment, leverage, holding period, and profit. In addition, we introduce a dummy variable that takes identifies short positions. We proxy investor risk-taking by investment, leverage, and total volatility. *Investment* is defined as the portfolio share invested in each trade when it is first executed. *Leverage* denotes the leverage of a trade, also when first executed. *Total volatility* is defined as leverage multiplied by stocks' volatility. It measures the total risk-taking of investors, as investors might refrain from leveraging their positions – which would indicate a decline in risk-taking – but instead invest in more volatile stocks (see, *e.g.*, Pelster, 2024). *Holding period* measures speculative behavior and is the time difference (in days) between the opening and closing of positions. Finally, the net profit proxy for investor financial welfare is the *holding period return (HPR)* of a position opened on a particular day. The HPR is the exact return of the position, based on the open rate and the closing rate of the trade, net of transaction costs.

We run (multidimensional) sentiment analyses on WSB user contributions to determine their emotional content. We employ *huggingface*, an open-source natural language processing and machine learning (ML) model hub and community platform that has close associations with Meta's AI division, Amazon Web Services, Microsoft, and Google AI. Specifically, we use the DistilRoBERTa-base model for emotion recognition.[19] The DistilRoBERTa-base model is trained on six diverse datasets and predicts Ekman's six emotions: joy (or happiness), surprise, anger, disgust, sadness, and fear (Ekman, Friesen, & Ellsworth, 1972). The underlying datasets are diverse and include among others labels for text from Twitter/X and Reddit. Note that these models only work on English language. The large majority of our posts (98%) are written in English. The algorithm assigns probability scores to each post, estimating to what degree the post is considered to reflect the emotions *joy*, *surprise*, *anger*, *disgust*, *sadness*, *fear*, and *neutral*. We label each post with its respective emotions by choosing the highest probability. Then, we count the posts for each ticker and day. Our measure is then the number of posts for each emotion per stock and day. As a counterpart to retail investor sentiment, we also employ analysts' *buy and sell recommendations*. In our analyses, for each stock, we include the difference between daily buy and sell recommendations, as reported in I/B/E/S.

We carefully control for the influence of potentially confounding effects with a series of control variables. With the purpose of clearly differentiating between attention received on social media and news, we include RavenPacks *event buzz*. It is a measure of abnormal news volume and computed as the number of standard deviations the daily event volume of a company is above or below its mean (Hafez et al., 2018). To control for the influence of public media outlets, we add a measure of average market sentiment – RavenPacks entity sentiment indicator (*ESI*) – to our list of variables. It is defined for any given asset as the average sentiment of events per day. Additionally, we add the companies' *market capitalization* because larger companies tend to receive more media attention (see, *e.g.*, Vozlyublennaia, 2014). Finally, we include the *price-to-book ratio*, a proxy for a value investing strategy. Although the strategy has performed quite poorly since 2007, it remains very present in the minds of young investors (Lev & Srivastava, 2022).

[19] https://huggingface.co/j-hartmann/emotion-english-distilroberta-base

Table 1
Top-10 WSB stocks.

	Ticker	Company	#WSB
1	gme	GameStop	514,943
2	tsla	Tesla	316,974
3	amzn	Amazon	80,289
4	msft	Apple	78,483
5	aapl	Microsoft	72,616
6	nvda	Nvidia	24,697
7	nflx	Tilray	21,276
8	pton	Netflix	19,758
9	snap	Peloton	17,745
10	wmt	Snapchat	16,291

The table shows the top-10 ticker/company name mentions on WSB from January 1, 2020, to February 12, 2021.

5.3. Data filter

The final dataset contains all stocks that have experienced sudden attention on social media. These meme stocks are defined to have more than 30 stock/ticker mentions and to have had at least one day of abnormal attention that is within the 95% quantile of nonzero days of abnormal attention. We repeat the analyses with 97.5% and 99%-quantiles, and the results remain qualitatively the same. As a result of our various data sources, data availability varies between analyses. For each analysis, we use the largest sample available.

5.4. Summary statistics

First, we study WSB user contributions. Fig. 1 shows that GameStop received most of the attention—by a wide margin. Approximately 34% of total ticker/company mentions can be attributed to Gamestop.

We show the top-10 companies based on WSB mentions in Table 1. Consistent with Fig. 1, the majority of posts has been dedicated to gme and tsla, the flagship meme stocks. The remaining top-10 constituents, mostly tech stocks, are familiar associates. One popular representative of meme culture, though, is missing from our list: AMC entertainment. This is simply for reasons of data availability. As our universe is limited to the trading grounds of the online brokerage service, which does not cover AMC, it is not included in our analyses.

Fig. 2 shows the industry sector compositions of the WSB portfolio. Stocks are distributed over various sectors, but attention on WSB is skewed toward companies from the technology and consumer cyclical sector.

We present an excerpt of the multidimensional sentiment analyses run on the WSB user contributions in Table 2. Table 2 shows the content of individual WSB user contributions and the associated sentiment for Activision Blizzard Inc. (atvi).

Table 4 shows descriptive statistics. Differences between the number of observations in WSB attention and the remaining variables in Panel A of Table 4 are due to public holidays, corporate events that cause individual stocks not to be traded (anymore), or data availability. An example of a corporate event during our sample period is the Aphria and Tilray merger. While the stock ticker continues to be mentioned on WSB, the stock can no longer be traded. Public holidays have the same effect on the number of observations. WSB attention is observed 24/7 and continues to generate observations when stock exchanges are closed. Differences in the number of observations between HPR, holding period, and others are because some positions in our dataset have been opened but were not yet closed at the time of data collection.

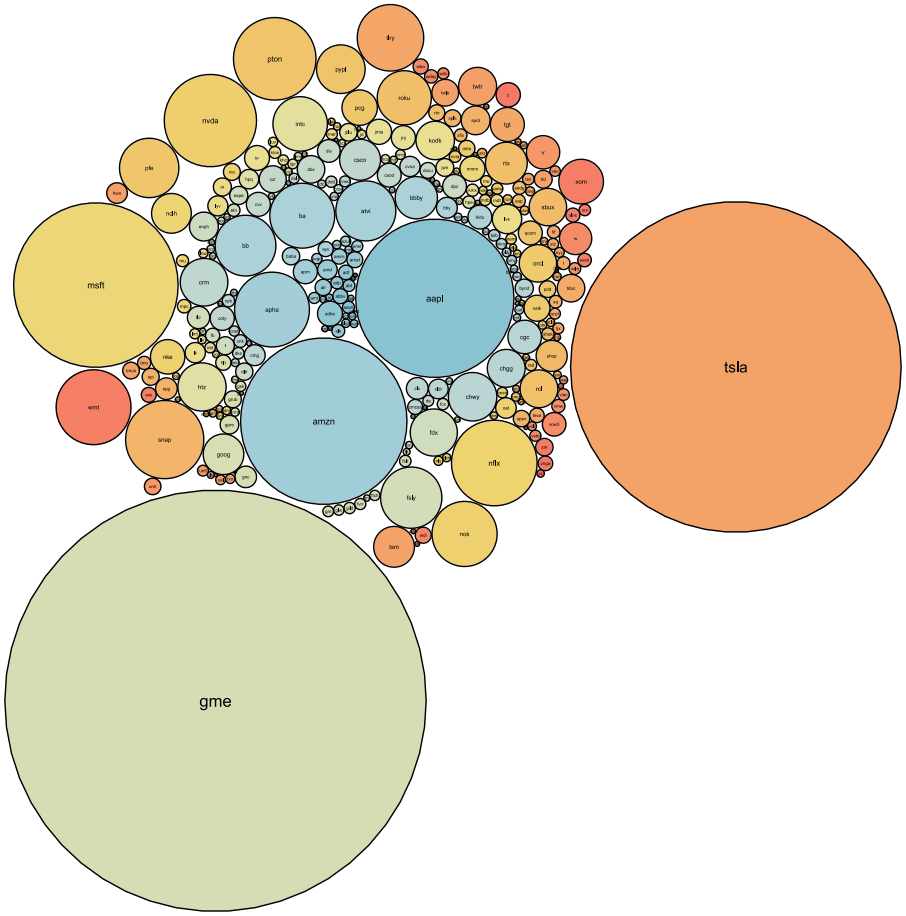

Fig. 1. Circular treemap of ticker/company mentions on WSB from January 1, 2020, to February 12, 2021. The sample consists of stocks that have experienced abnormal attention on WSB.

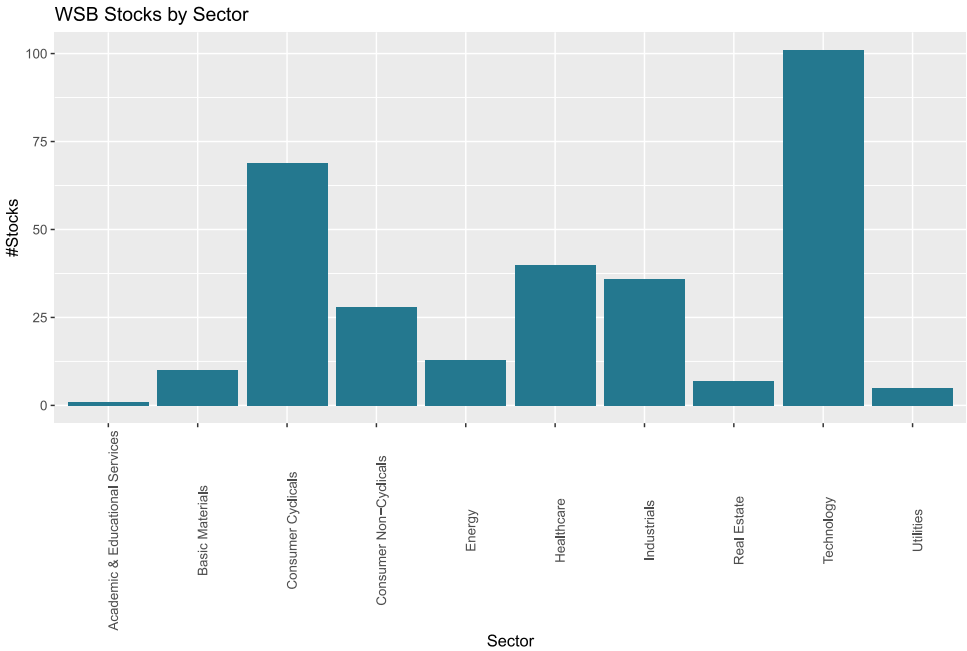

Fig. 2. This figure shows the industry sector composition of WSB stocks.

Table 2
Emotion recognition.

Text	Sentiment
i think atvi is a good play	joy
i think the atvi is solid. good luck to us	joy
i love how atvi is resilient.	joy
so how come atvi is surviving today	surprise
it amuses me that atvi is just sitting there up almost 2%. wtf.	surprise
wow even my atvi leaps are starting to come back.	surprise
i need atvi to take a shit	anger
are my atvi calls fucked?	anger
couldn't get a fill and now atvi is running. fucking garbage	anger
if you believe the atvi stock, it is dogshit	disgust
well, atvi is a steaming pile of wombat crap.	disgust
benzinga can suck a fat dick, they stole atvi's thunder	disgust
everyday i look at atvi and am throughly depressed	sadness
i already know my atvi is gonna sell off by open and it's gonna be painful to watch	sadness
as an atvi call holder i am an extremely unhappy individual	sadness
i have $85c atvi 8/14 and i'm kinda scared	fear
if atvi kisses 53.5 ill panic	fear
kinda scared to hold atvi through er	fear

This table shows examples of the emotion recognition results for the WSB posts. All posts in the table are associated with atvi.

Table 3
A selection of WSB posts with no informational content.

Post	Ticker	Time	Author
the hour of flws is upon us	flws	2021-12-14 19:53:15	Pojihut
is flws the next meme stonk lol?	flws	2021-05-03 05:47:42	Jonomac420
what about flws my guy??	flws	2021-05-01 00:08:02	Baconstrips96
i'm feeling bullish on flws there must be something in the air	flws	2021-02-11 21:37:49	zutroyapple
mother's day flws may 9 is mother's day. don't forget to buy her some flowers flws!!! i like the stonk!	flws	2021-04-29 00:27:18	Mullinax
i guarantee there is atleast one god tier autist out there that bought ddd calls just because of the ticker	ddd	2021-01-08 15:20:07	IcyTitle1
made 300$ in 20 mins on the last part of that ddd run an sold before the drop. am not much, but its honest work	ddd	2021-01-08 17:31:46	AngloWaxson
holy cow $mmm is looking parabolic	mmm	2021-02-18 18:35:35	HeAbides
buy calls $mmm i see sanitizer on the rise.	mmm	2020-01-24 23:20:51	abiech
a year ago i posted a thread in r/stocks. "why will not my 3 m shares go up?" then i found this sub and everything changed.	mmm	2021-02-26 13:05:17	jaboyles

This table shows examples of non-informative posts on WSB.

Table 4 shows that the average daily return of meme stocks is around zero, while investors on the trading platform gain, on average, 3.1% per trade. Our measure of attention, abnormal WSB mentions, has a mean of almost 2 and is slightly right-skewed. The average daily buy–sell imbalance is positive, and each stock is traded approximately 590 times per day. The majority of trades (92.7%) are long positions, and 7.3% are short positions. Average investment per stock is 16% of portfolio value, and investors lever their positions, on average, with a factor of 2.1. The holding period per stock is almost 11 days. However, this number is influenced by some very long investments (the .75 quantile is at 8 days).

6. Empirical analyses

We estimate a set of standard ordinary least squares (OLS) models. Standard errors are double-clustered, adjusting for correlations across firm and time. We also include firm and day-of-the week effects (fixed effects) to account for unobservable (and observable) firm and time patterns.

6.1. WSB attention and other attention measures

First, we briefly benchmark WSB against other market-wide sentiment measures in the literature to investigate if WSB attention is different from other attention measures. We exploit ASVI (Da et al., 2011) as well as the social attention signal by Cookson et al. (2024) that aggregates social media attention from the three major platforms Twitter, StockTwits, and Seeking Alpha.

We first study Pearson correlation coefficients across the three attention measures. in the Appendix (Panel A) indicates that attention measures are indeed positively correlated; however, bivariate correlations are around 20% across measures. Second, we estimate a simple linear regression model on WSB attention. While we find that indeed both ASVI and the social attention signal significantly correlate with WSB attention (see Panel B in), we also observe that the R^2 amounts to only 14%. Thus, a large fraction of WSB attention remains unexplained.

Overall, the evidence suggests that WSB attention is different from other market-wide attention measures in the literature.

6.2. WSB attention and retail trading at the micro level

Before we turn to a formal analysis of our hypotheses, we briefly investigate the predictive power of WSB attention for buy–sell imbalances in our trade data. Doing so, we provide evidence that the micro-level trading data corresponds with the macro-level results in the literature (Avila et al., 2023; Bradley et al., 2024; Cookson et al., 2024; Hu et al., 2023). Adding to the literature, the micro-level data allow us to shed light on the short-selling activities by retail investors.

Table 4
Summary statistics.

Statistic	N	Mean	St. Dev.	25	Median	75
Panel A: Market data						
Return	83,973	0.001	0.048	−0.015	0.001	0.018
GARCH volatility	83,973	0.036	0.028	0.020	0.030	0.043
Volume	83,969	13.810	1.319	12.960	13.768	14.646
WSB attention	88,651	4.114	352.631	−0.667	−0.133	0.300
Event buzz	74,544	0.320	1.131	−0.390	−0.041	0.643
ESI	74,546	0.156	0.255	0.000	0.185	0.335
Market capitalization	83,973	69,706.810	184,665.300	4,712.825	16,847.620	55,983.370
Price-to-book	83,691	6.764	68.837	1.502	4.066	10.441
Panel B: Individual trade data						
BSI	83,940	3.802	28.230	−10.000	2.413	18.660
Trading intensity	83,940	587	3,390.931	19	68	257
Long position	26,245,571	0.927	0.261	1	1	1
Investment	26,245,571	15.581	23.421	2.020	6.100	17.320
Leverage	26,245,571	2.140	2.218	1	1	2
Holding period	22,875,796	10.710	27.226	0.032	1.036	7.826
HPR	22,875,796	3.052	42.581	−1.673	0.615	5.250

This table shows summary statistics of the variables used in the analyses. The variables include daily stock returns, volatility, stock market volume (daily number of traded shares), abnormal attention on WSB (number of WSB posts for a specific stock on a certain date minus its 30-day moving average), event buzz (number of standard deviations the daily event volume of a company is above or below its mean), sentiment (average sentiment of events per day), market capitalization, and the stock's price-to-book ratio. The brokerage data contain the variables buy–sell imbalance, measured as the number of long minus short to total trades, and trading intensity, which is the number of retail trades per day. Additionally, the data provide us with information on the number of long positions, investment weights (portfolio share invested into each trade), leverage, holding periods (time difference in days between opening and closing of positions), and individual holding period returns. Data are available from January 1, 2020, to February 12, 2021.

Barber and Odean (2008) were the first to show that individual investors are net buyers of attention-grabbing stocks. Due to limited cognitive abilities, investors tend to buy stocks that grab their attention when deciding where to invest. Of course, the search problem for new positions also exists for short sellers, and retail investors sell short more frequently in recent years. While Barber and Odean (2008) report that short sales reflect only 0.3% of trades in their sample from the 1980s and 1990s, Kelley and Tetlock (2017) find that approximately 6% of the dollar volume of all executed orders to be short sales. In line with this number, we find approximately 7% of trades in our dataset to be short sales. Consequently, we do not limit our analyses to the buying and selling behavior of traditional long investors, but consider different buy–sell imbalance variants. In particular, we compute the buy–sell imbalance based on (i) both long and short positions, (ii) long positions, and (iii) short positions. We estimate the following model:

$$y_{i,t} = \alpha + \beta\ attention_{i,t-1} + \sum_{j=1}^{J} \gamma_j\ stock\ controls_{ij,t-1} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \eta_n\ stock_{ni} + \epsilon_{i,t}, \tag{1}$$

where $y_{i,t}$ are variants of the buy–sell imbalance, computed based on long, short and both long and short positions for firm i during day t. The variable of interest is WSB $attention_{i,t-1}$. Stock controls include lagged *buy–sell imbalance*, *event buzz*, *sentiment*, *return*, *market capitalization*, and *price-to-book ratio* for firm i during day $t-1$. We summarize the results in Table 5.

We find a positive and significant coefficient for our attention measure on buy–sell imbalance, which is computed based on long and short trades (Column 1). When calculating the measure on long and short trades individually, one can observe that not only are traditional long trades affected by WSB attention (Column 2), but the same holds for short trades (Column 3). Note that the difference in signs of Column 2 and Column 3 is because long investors have to choose from a wide variety of stocks when *buying*, while short investors choose from a wide variety of stocks when *selling*.

Altogether, the results are in line with the literature (Avila et al., 2023; Bradley et al., 2024; Cookson et al., 2024; Hu et al., 2023) and indicate that abnormal WSB attention is a predictor of retail investor net buying. Adding to the literature, we document that attention also affects short-selling activities.

6.3. WSB attention and investor risk-taking

Next, we begin the formal analysis of our hypotheses and turn to investors' risk-taking. Hypothesis 1 proposes that WSB attention increases investors' risk-taking. Therefore, we assess the predictive power of attention on trade characteristics. We regress *WSB attention* on *leverage*, *volatility*, *total volatility*, *holding period*, and *investment*. Formally, we estimate the following model:

$$y_{il,t} = \alpha + \beta\ attention_{i,t-1} + \sum_{j=1}^{J} \gamma_j\ stock\ controls_{ij,t-1} + \sum_{k=1}^{K} \delta_k\ transaction\ controls_{lk,t} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \eta_n\ stock_{ni} + \epsilon_{il,t}, \tag{2}$$

where $y_{il,t}$ is either *leverage*, *volatility*, *total volatility*, *holding period*, or *investment* weight for position l opened on day t. *Total volatility* is the stock's volatility multiplied by leverage. The variable of interest is $WSB\ attention_{i,t-1}$ for firm i during day $t-1$. Stock controls include *buy–sell imbalances*, *event buzz*, *sentiment*, *return*, *market capitalization*, and *price-to-book ratio* for firm i during day $t-1$. Transaction controls are *holding period* and a *long/short indicator* for position l. We summarize the results Table 6.

Overall, the results provide partial support for Hypothesis 1. We find that attention negatively predicts *leverage* (Column 1), is a positive predictor of volatility (Column 2), and has no effect on the position's total volatility (Column 3). This evidence suggests that investors swap volatility for leverage. When investors' WSB attention increases, they do not increase their risk-taking by using highly levered positions but instead engage in unlevered trades that favor stocks with high volatility. Also, we find that attention is a positive predictor of investment (Column 4). Finally, we observe a coefficient close to zero for *holding period* (Column 5). These results indicate that investors increase their

Table 5
The influence of attention on retail buy–sell imbalances.

	Dependent variables		
	BSI (1)	BSI long (2)	BSI short (3)
WSB attention$_{t-1}$	0.0004**	0.0004**	−0.0003***
	(2.4469)	(2.2183)	(−3.8025)
BSI$_{t-1}$	0.1236***		
	(8.9272)		
BSI long$_{t-1}$		0.1338***	
		(9.9094)	
BSI short$_{t-1}$			−0.0945***
			(−16.9130)
Event buzz$_{t-1}$	0.2914***	0.1966**	0.4548***
	(3.0251)	(1.9961)	(2.9423)
ESI$_{t-1}$	0.1709	−0.0433	−0.5666
	(0.3148)	(−0.0776)	(−0.8334)
Return$_{t-1}$	−11.5402**	−15.4987***	0.1000
	(−2.1037)	(−2.6820)	(0.0196)
Market cap$_{t-1}$	0.00001***	0.00001***	−0.000000
	(3.9636)	(4.0171)	(−0.0153)
Price to book$_{t-1}$	−0.0025**	−0.0014	0.0017
	(−1.9938)	(−1.0314)	(0.8511)
Stock effects	Yes	Yes	Yes
Date effects	Yes	Yes	Yes
Stocks	288	288	288
Observations	68,009	68,009	68,009
Adjusted R^2	0.0857	0.0942	0.0179

*p<0.1; **p<0.05; ***p<0.01.
This table shows the results of multivariate panel regressions. The dependent variables are daily buy–sell imbalances, which are computed based on both long and short positions (Column 1), only long positions (Column 2) and only short positions (Column 3). The regressors are WSB abnormal attention (number of WSB posts for a specific stock on a certain date minus its 30-day moving average), buy–sell imbalances (long minus short to total positions), event buzz (number of standard deviations the daily event volume of a company is above or below its mean), sentiment (average sentiment of events per day), stock return, market capitalization, and price-to-book ratio. We also include individual stock and date-fixed effects. Standard errors are double-clustered (firm and time). *t*-statistics are in parentheses.

Table 6
The influence of attention on investor risk-taking.

	Dependent variables				
	Leverage (1)	Volatility (2)	Total volatility (3)	Investment (4)	Holding period (5)
WSB attention	−0.00001***	0.000003***	0.000001	0.0001***	0.00001
	(−3.0552)	(3.5663)	(0.8767)	(3.8691)	(0.7873)
BSI	0.0001	0.00003	−0.0001	−0.0079**	0.0379***
	(0.2674)	(0.3320)	(−0.7989)	(−2.1404)	(5.7790)
Long position	−1.3664***	0.0017	−0.0803***	−6.8530***	8.7911***
	(−24.7997)	(0.6474)	(−14.2916)	(−9.2075)	(6.8513)
Event buzz	0.0030	0.0029**	0.0061**	0.2849***	−0.1982***
	(0.2945)	(2.5329)	(2.1497)	(5.0774)	(−4.3269)
ESI	0.0179	−0.0190*	−0.0298***	0.1136	0.4519
	(0.3174)	(−1.9503)	(−2.7402)	(0.4046)	(1.0405)
Return	−0.0916	−0.0146	−0.0328	1.7829*	−1.0248
	(−0.5555)	(−0.4731)	(−0.6167)	(1.7362)	(−1.2504)
Market cap	−0.000001**	−0.0000	−0.0000	0.000001	0.00001***
	(−2.2143)	(−0.3347)	(−0.0737)	(1.1597)	(3.2521)
Price to book	−0.0004**	0.00001	−0.00003**	−0.0014	−0.0007
	(−2.0600)	(0.4555)	(−2.3616)	(−0.6397)	(−0.2752)
Stock effects	Yes	Yes	Yes	Yes	Yes
Date effects	Yes	Yes	Yes	Yes	Yes
Stocks	288	288	288	288	288
Observations	21,760,640	21,760,640	21,760,640	21,760,640	21,760,640
Adjusted R^2	0.2287	0.7435	0.3837	0.0547	0.0639

*p<0.1; **p<0.05; ***p<0.01.
This table shows the results of multivariate panel regressions. The dependent variables include leverage, volatility, total volatility (leverage multiplied by the stock's volatility), investment weights (portfolio share invested in each trade), and holding period (time difference in days between the opening and closing of positions). Regressors include abnormal WSB attention (number of WSB posts for a specific stock on a certain date minus its 30-day moving average), buy–sell imbalances (long minus short to total positions), holding period, long position indicator (dummy variable that takes value 1 for a long position and 0 for a short position), event buzz (number of standard deviations the daily event volume of a company is above or below its mean), sentiment (average sentiment of events per day), stock return, market capitalization, and price-to-book ratio. We also include individual stock and day-of-the-week effects. Regression coefficients are presented with *t*-statistics in parentheses. Standard errors are double-clustered (firm and time).

risk-taking but in a less apparent manner than expected. Investors refrain from recklessly leveraging their positions and instead invest in more volatile stocks, trade more frequently, and increase their position sizes.

While we do not find leverage to increase in attention, attention nevertheless increases investor risk-taking. The reason why meme investors increase their risk-taking not through higher leverage but by investing in more volatile stocks and engaging in speculative behavior could be due to the underlying sources of attention. Ben-Rephael, Da, and Israelsen (2017), for example, show that institutional attention, which is measured by news reading activity on Bloomberg terminals, is different from retail attention, as proxied by Google searches. The paper's conclusion "it depends on where you search", might also be the explanation for the discrepancy of our results in relation to previous literature. In the case of Arnold et al. (2022), attention arrives via push messages directly from the brokerage service sent privately to the users account. In contrast, our measure of attention has its origins in social media, is publicly available, can be commented on and stems from peers. It could simply be that retail investors are keen on participating in whatever is fashionable at that moment ("fear of missing out") but do not sufficiently trust their peers to lever up.

6.4. WSB attention and retail holding-period returns

In H2, we hypothesize that abnormal attention on WSB spurs uninformed trading and harms investor welfare. This implies that retail buy–sell imbalances of executed trades cannot positively predict holding period returns, *i.e.*, that retail trading is not informative. Thus, we first investigate the predictive power of retail buy–sell imbalances for the holding period returns and expect the coefficient of BSI to be either not significantly different from zero or significantly negative (Eq. (3)). If there were no effect, we would conclude that retail buy–sell imbalances are uninformative. If the effect were negative, we would reason that retail investors make suboptimal trading decisions and trade at temporarily disadvantageous prices, because arbitrageurs require compensation (Boehmer et al., 2021). If the coefficient were positive, one could infer, among other things, that retail traders are informed and that they have some kind of information advantage that is only incorporated into prices some time after their trades have occurred.

We estimate the following model:

$$HPR_{il,t} = \alpha + \beta\ BSI_{i,t-1} + \sum_{j=1}^{J} \gamma_j\ stock\ controls_{ij,t-1} + \sum_{k=1}^{K} \delta_k\ transaction\ controls_{lk,t} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \eta_n\ stock_{ni} + \epsilon_{il,t}, \tag{3}$$

where $HPR_{il,t}$ is the holding period return for position l that was created during day t. The variable of interest is $BSI_{i,t-1}$ for firm i during day $t-1$. Stock controls include *event buzz*, *sentiment*, *return*, *market capitalization*, and *price-to-book ratio* for firm i during day $t-1$. Transaction controls are *holding period* and a *long/short indicator* for position l (Arnold et al., 2022).

To analyze whether uninformed trading is spurred by investors collectively trading on the noisy signal attention, we first regress attention on buy–sell imbalances:

$$BSI_{i,t} = \alpha + \beta\ attention_{i,t-1} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \eta_n\ stock_{ni} + \eta_{i,t}. \tag{4}$$

We then use the fitted values of Eq. (4) to capture the part of retail trading that can be explained by attention and regress it on investor holding period returns:

$$HPR_{il,t} = \alpha + \beta\ \widehat{BSI}_{il,t-1} + \sum_{j=1}^{J} \gamma_j\ stock\ controls_{ij,t-1} + \sum_{k=1}^{K} \delta_k\ transaction\ controls_{lk,t} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \eta_n\ stock_{ni} + \epsilon_{ilt}. \tag{5}$$

Note that the estimation of Eq. (4) is based on aggregate market outcomes, while the estimation of Eq. (5) is based on individual investors' decisions. As a result, they have different sample sizes.

We also consider the counterfactual, the residual of Eq. (4), to investigate those patterns in retail trading that cannot be accounted for by attention:

$$HPR_{il,t} = \alpha + \beta\ \eta_{l,t-1} + \sum_{j=1}^{J} \gamma_j\ stock\ controls_{ij,t-1} + \sum_{k=1}^{K} \delta_k\ transaction\ controls_{lk,t} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \theta_n\ stock_{ni} + \epsilon_{il,t}. \tag{6}$$

Control variables remain the same as in Eq. (3). We summarize regression results in Table 7.

Table 7 shows a coefficient of −0.03 for BSI that is significantly different from zero (Column 1, t-statistic of 1.97). This indicates that the trading patterns of meme-stock traders are not informative. In Columns 3 and 4, we decompose BSI into retail trading patterns that can be explained by attention (Column 3) and those that cannot (Column 4). We find a negative (−1.44) and highly significant (t-value of 7.97) coefficient on *BSI attention*, meaning that meme investors make costly mistakes when trading on attention. The coefficient of *BSI other* is not significantly different from zero.

In our next step, we dig deeper and consider the impact of WSB attention on individual investor welfare. To this end, we estimate the following regression model:

$$HPR_{il,t} = \alpha + \beta\ attention_{i,t-1} + \sum_{j=1}^{J} \gamma_j\ stock\ controls_{ij,t-1} + \sum_{k=1}^{K} \delta_k\ transaction\ controls_{lk,t} + \sum_{m=1}^{M-1} \zeta_m\ date_{m,t} + \sum_{n=1}^{N-1} \eta_n\ stock_{ni} + \epsilon_{il,t}, \tag{7}$$

where $HPR_{il,t}$ is the holding period return for position l opened on day t. The variable of interest is WSB $attention_{i,t-1}$ for firm i during day $t-1$. Stock controls include *buy–sell imbalances*, *event buzz*, *sentiment*, *return*, *market capitalization*, and *price-to-book ratio*. Transaction controls are *holding period* and a *long/short indicator* for position l. We run this regression on three different samples, based on empirical evidence that suggests that retail short sellers have an informational advantage, even compared to analysts (Boehmer et al., 2021; Kelley & Tetlock, 2017). This informational advantage might simplify their search problem and make them less vulnerable to attention. We summarize the results in Table 8, Panel A. Column (1) includes long and short traders alike, Column (2) includes those investors that have never opened a short position. Because Column (2) only includes long positions per definition, the control variable long/short indicator is not needed. Column (3) includes only short sellers, *i.e.*, everyone that has at least once made a short sale.

Panel A of Table 8 shows that attention on WSB negatively predicts holding period returns, irrespective of the underlying sample. The negative correlation suggests a direct correlation between WSB attention and investor financial welfare. Greater abnormal attention on WSB a day before the investment is related to worse outcomes. These results align with the finding that WSB attention is able to predict the trading patterns of not only long but also short trades, as reported in Table 5. The informational advantage of short sellers, as reported in, for example, Kelley and Tetlock (2017), does not appear in the context of social media attention on WSB. Of course, factors other than WSB attention prior to opening a position might influence profits over the holding period. Nonetheless, we believe this evidence to be at least suggestive of the notion that attention-induced investments result in negative profits.

Of course, one might argue that not all stocks are alike. Specifically, stocks from companies that receive less (media) coverage may be less

Table 7
Attention spurs uninformed trading in meme investors.

	Dependent variables			
	HPR	BSI	HPR	
	(1)	(2)	(3)	(4)
BSI_{t-1}	−0.0254*			
	(−1.9658)			
WSB $attention_{t-1}$		0.0006***		
		(2.8622)		
BSI attention			−1.4386***	
			(−7.9691)	
BSI other				−0.0169
				(−1.0376)
Holding period	0.3248***		0.3249***	0.3249***
	(3.8815)		(3.8848)	(3.8812)
Long position	5.1054***		5.3201***	5.1201***
	(3.8179)		(3.9803)	(3.8472)
Event $buzz_{t-1}$	−0.5961**		−0.2532*	−0.6412**
	(−2.4190)		(−1.7396)	(−2.3301)
ESI_{t-1}	−1.8038		−0.7751	−1.7845
	(−1.6129)		(−0.5821)	(−1.5787)
$Return_{t-1}$	−1.8193		−10.1740**	−1.3378
	(−0.2720)		(−2.0179)	(−0.2007)
Market cap_{t-1}	−0.00001***		−0.00001***	−0.00001***
	(−3.7764)		(−3.8624)	(−3.8400)
Price to $book_{t-1}$	−0.0128		−0.0070	−0.0129
	(−1.2699)		(−1.2266)	(−1.2748)
Stock effects	Yes	Yes	Yes	Yes
Date effects	Yes	Yes	Yes	Yes
Stocks	288	288	288	288
Observations	21,760,499	79,616	21,760,499	21,760,499
Adjusted R^2	0.0609	0.0675	0.0662	0.0609

*p<0.1; **p<0.05; ***p<0.01.
This table shows the results of multivariate panel regressions. The dependent variables are holding period returns and buy–sell imbalances (long minus short to total positions). The regressors are buy–sell imbalances, WSB abnormal attention (number of WSB posts for a specific stock on a certain date minus its 30-day moving average), and fitted values and residuals from Column (2). Transaction controls are holding period and a long/short indicator. Stock controls include event buzz (number of standard deviations the daily event volume of a company is above or below its mean), sentiment (average sentiment of events per day), stock return, market capitalization, and price-to-book ratio. We also include individual stock and date-fixed effects. Regression coefficients are presented with *t*-statistics in parentheses. Standard errors are double-clustered (firm and time).

informationally efficient. Consequently, social media users might have a better chance to benefit from paying attention to those stocks. To investigate whether our finding that WSB attention extends to stocks which receive less attention in other sources, we repeat our analysis but exclude the largest stocks and those with the largest analyst coverage. We use size as a proxy for coverage by the media as the largest firms are typically most frequently covered in the media. We report the results of the analyses in Panels B and C of Table 8, respectively. For both subsamples, we observe very similar results, indicating that even for firms that are less heavily covered, investors who trade on WSB attention tend to lose money, on average.

Next, one may argue that information obtained from WSB is exploited immediately. While we consider HPR and thus the exact time of opening and closing of positions, our measure of WSB is thus far measured at the daily level. To obtain a more granular view of the relationship between WSB attention and HPR, we now consider WSB attention at the hourly level. Then, we regress HPR on our hourly measure for WSB attention, lagged by one hour, while controlling for lagged hourly BSI and including stock and hour-fixed effects. We summarize our results in Panel D of Table 8. We observe a significantly negative coefficient of hourly WSB attention on HPR. While the coefficient is much larger than in the models using daily WSB attention, the marginal effect is actually somewhat smaller in the model using the hourly WSB measure. This is because, obviously, the number of WSB posts per hour is much smaller than the number of posts per day, and the standard deviation is also of smaller magnitude. Overall, the intraday level data supports our findings.

Thus far, we have mainly focused on the effect of attention on the opening of positions, *i.e.*, we have related yesterday's attention to the trade characteristics and outcomes of positions that were opened today. Now, we follow a more comprehensive approach by considering all *attention trades*. We define attention trades as trades – which can either be the opening or closing of positions – with a high probability of being executed *due* to high attention on WSB. In particular, we investigate whether investors suffer financial losses from engaging in attention trades and whether these trades are characterized by speculative behavior.

To this end, we restrict our data sample to stocks that are within the 99% quantile of nonzero days of abnormal attention, which yields a total of 76 stocks. Within these stocks, we identify attention trades at the local maxima (peaks) subject to the constraint that the minimum peak height is at least within the 99%-quantile of a stock's number of daily WSB posts. The minimum distance between peaks is set to 30 days. For the 76 stocks in our sample, we identify 121 attention shocks. In our analyses, we then examine the average number of WSB user contributions, holding period returns at time of purchase and sale, absolute number of buy–sell relations, buy–sell imbalances, individual investor characteristics, and trading intensity on high-attention days (peaks) and the immediate days before and after.

Fig. 3 illustrates the methodology. It shows the daily WSB posts of Activision Blizzard Inc. (atvi) from May to August 2020. We identify peaks (squares) that are within the 99%-quantile of atvis WSB mentions and examine days t−4 to t+4 (shaded area). The high number of posts on atvi in both these instances relates to the publishing of the companies' quarterly earnings reports.

Table 9 reports averages from t−4 to t+4 relative to the peak for our variables of interest across the 76 high-attention stocks considered in

Table 8
The influence of attention on investor welfare.

Panel A: Full sample			
	Dependent variables		
	HPR (1)	HPR long buyer (2)	HPR short seller (3)
WSB attention$_{t-1}$	−0.0009***	−0.0011***	−0.0008***
	(−8.0100)	(−5.8583)	(−9.3209)
BSI$_{t-1}$	−0.0060	−0.0142	−0.0020
	(−0.4531)	(−0.7050)	(−0.2011)
Holding period	0.3250***	0.3184***	0.3395***
	(3.8853)	(3.8120)	(3.9084)
Long position	5.3267***		5.0381***
	(3.9968)		(4.1355)
Event buzz$_{t-1}$	−0.2398*	−0.3819**	−0.1893
	(−1.7471)	(−2.0208)	(−1.6272)
ESI$_{t-1}$	−0.7909	−1.0236	−0.8696
	(−0.5904)	(−0.4202)	(−1.0013)
Return$_{t-1}$	−10.2296**	−12.3868	−8.1813**
	(−2.0200)	(−1.6327)	(−2.3782)
Market cap$_{t-1}$	−0.00001***	−0.00002***	−0.00001***
	(−3.8512)	(−2.8791)	(−4.3543)
Price to book$_{t-1}$	−0.0071	−0.0148	−0.0031
	(−1.2342)	(−1.5168)	(−0.8566)
Stock effects	Yes	Yes	Yes
Date effects	Yes	Yes	Yes
Stocks	288	288	288
Observations	21,760,640	7,951,433	13,809,207
Adjusted R^2	0.0662	0.1040	0.0478
Panel B: Exclude largest firms			
	Dependent variables		
	HPR (1)	HPR long buyer (2)	HPR short seller (3)
WSB attention$_{t-1}$	−0.0008***	−0.0009***	−0.0007***
	(−6.6295)	(−5.1366)	(−7.5266)
Controls	Yes	Yes	Yes
Stock effects	Yes	Yes	Yes
Date effects	Yes	Yes	Yes
Stocks	237	237	237
Observations	10,205,322	4,747,602	5,457,720
Adjusted R^2	0.0672	0.1018	0.0448
Panel C: Exclude high-analyst-coverage firms			
	Dependent variables		
	HPR (1)	HPR long buyer (2)	HPR short seller (3)
WSB attention$_{t-1}$	−0.0008***	−0.0009***	−0.0007***
	(−5.5913)	(−4.7485)	(−5.9314)
Controls	Yes	Yes	Yes
Stock effects	Yes	Yes	Yes
Date effects	Yes	Yes	Yes
Stocks	256	256	256
Observations	8,384,421	4,099,449	4,284,972
Adjusted R^2	0.0588	0.0812	0.0408
Panel D: Intraday WSB attention			
	Dependent variables		
	HPR (1)	HPR long buyer (2)	HPR short seller (3)
WSB attention$_{t-1}$	−0.4239**	−0.3883*	−0.4328**
	(−2.2867)	(−1.8091)	(−2.4152)
Controls	Yes	Yes	Yes
Stock effects	Yes	Yes	Yes
Hour effects	Yes	Yes	Yes
Stocks	51	51	51
Observations	12,856,760	4,599,822	8,256,938
Adjusted R^2	0.0217	0.0442	0.0147

*p<0.1; **p<0.05; ***p<0.01.
This table shows the results of a multivariate panel regression. The dependent variables are holding period returns. Panel A contains the full sample; Panel B excludes the largest firms; Panel C excludes firms with the highest analyst coverage. Regressors in all panels are abnormal WSB attention (number of WSB posts for a specific stock on a certain date minus its 30-day moving average), buy–sell imbalances (long minus short to total positions), holding period (time difference in days between opening and closing of positions), long position indicator (dummy variable that takes value 1 for a long position and 0 for a short position), event buzz (number of standard deviations the daily event volume of a company is above or below its mean), sentiment (average sentiment of events per day), stock return, market capitalization, and price-to-book ratio. We also include individual stock and date-fixed effects. Regression coefficients are presented with *t*-statistics in parentheses. Standard errors are double-clustered (firm and time).

this analysis. The holding period returns divided by time of purchase (HPR open) are positive shortly before attention peaks, and investors gain, on average, 13% of portfolio value in t−4 and t−3. However, returns become negative in t−2 and continue being negative until t+4. Holding period returns divided by time of sale show quite the opposite picture. They are positive until the peak day – up to 32% – and become negative shortly after the peak in t+2. Absolute net buying #Buy-#Sell, buy–sell imbalances, and trading intensity increase until attention peaks then decrease in value. Averages across the variables provide further evidence of a relation between WSB attention and retail

Table 9
Attention trades.

Days	#WSB	HPR open	HPR close	#Buy-#Sell	BSI	Trading intensity
$\bar{x}$	49.93	2.82	2.82	79.10	375.34	1244.89
−4	94.85	13.60	14.91	334.88	668.50	3099.71
−3	89.98	13.02	10.54	294.10	878.85	3720.59
−2	181.17	−2.77	32.22	715.03	1364.73	3497.02
−1	677.74	−6.04	14.39	1382.65	1671.12	6285.49
Peak	1040.00	−8.52	6.78	1702.59	1782.94	8882.59
+1	735.40	−9.72	0.54	606.21	468.51	7483.41
+2	586.02	−5.24	−6.81	648.79	402.38	5274.48
+3	288.48	−2.33	−6.07	86.53	−0.52	3798.60
+4	117.28	−1.43	−13.62	−101.81	267.05	2477.45

This table shows descriptive statistics for attention trades. It shows averages for the number of daily WSB postings, holding period returns for positions opened on a given day (HPR open) and closed on a given day (HPR close), absolute differences between long and short positions, buy–sell imbalances, and trading intensity. Averages ($\bar{x}$) are based on the overall period (01/2020-02/2021), which includes high attention (peak) dates.

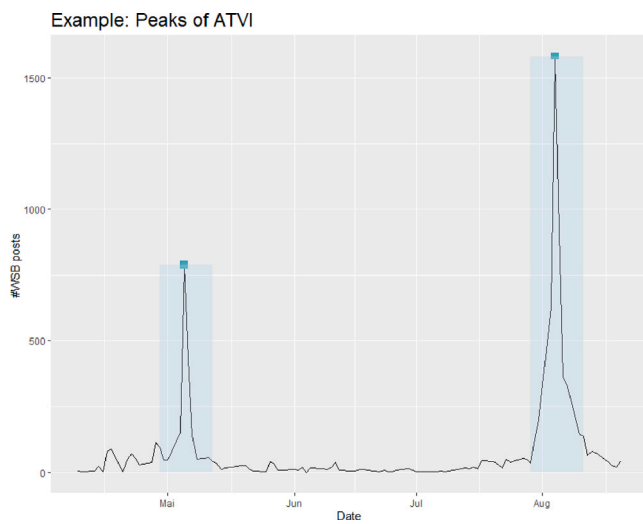

Fig. 3. This figure shows the methodology used to identify attention trades. This particular time series is of atvi.

trading behavior, as average values considerably deviate from values observed when WSB attention is high.

In the following, we define attention trades to be trades that occur between $t-1$ and $t+2$ in Table 9, because these days, on average, show the highest levels of WSB posts across stocks. This information is now linked to the individual trade-level data. Specifically, we identify (1) positions that have been opened and closed during periods of high attention, (2) those that have only been opened (excludes (1)), (3) all positions that have been opened, which is the sum of (1) and (2), and (4) those that have been opened before but closed during high-attention periods.

Table 10 indicates that investors who invested during periods of high attention lose money. On average, returns are −7.66%. They lose significantly more money (−14.74%) when they keep their trades open beyond times of high attention. However, if the trades are closed before attention fades, investors still lose but approximately 10 percentage points less. If they are not opening but closing their positions, investors on average generate a net profit of almost 39%. Note that only 14% of all trades that occurred in high-attention stocks in periods of high attention were closed. The ratio of investors that opened and closed their positions during periods of high attention to those that kept their positions open is approximately 60/40.

To provide further evidence for the observation that the results are indeed because of attention and not coincidental, we additionally analyze all variables in different contexts. First, we consider all stocks in the broker universe except the high-attention stocks on the dates that high attention occurs (5)-(6). With this subset, we aim to rule out that the results are driven by isolated events on specific dates that affect all stocks equally. The second control group consists of all attention stocks (those that at least once have experienced high attention) on all dates except the attention dates (7). This rules out stock-specific concerns. Third, we include the averages of all tradable stocks in the brokerage universe across all dates (includes high attention trades) (8) for general robustness.

Comparing the profits of attention trades to the other groups, we find remarkable differences. HPRs seem to be overall positive in a range from 2.32% to 3.46% across all stocks on all dates, while they are highest for attention stocks on all dates except high-attention dates. The difference between attention trades and the comparison groups is obvious and provides further evidence in favor of Hypothesis 2. Attention on WSB seems to harm investor welfare, because investors who open positions on these days, on average, suffer from double-digit losses. However, for the minority of investors, who have held stocks in their portfolios before they became high-attention stocks, the frenzies are very lucrative.

We have shown that investors who engage in attention trades suffer financial losses. Now, we investigate whether attention trades are characterized by speculative behavior. Table 8 shows that attention is a negative predictor of leverage. This result corresponds with Table 10 for those investors who open but do not close their trades for high-attention stocks. The average leverage is 1.52, which is considerably less than the average of 2.15 for attention trades. However, attention day-traders (1) are more risk-seeking and leverage their positions, on average, by 2.61. While this might be evidence for different subgroups of investors, in total, we do not observe a higher level of leverage for high-attention trades (3) than other groups who apply leverage somewhere between 1.92 (5) and 2.45 (7). However, when considering the total volatility of positions, we find support for the results from Table 10, which show that investors swap leverage for high-volatility stocks. The total volatility of positions of high-attention stocks opened during high-attention days is approximately twice as high as in all other scenarios.

Furthermore, attention trades are associated with higher investments of 19.81% compared to approximately 14% of portfolio value for the benchmark groups. Note that investments in attention stocks with, on average, 17.61% in general seem to be slightly higher than the remaining groups. Shorter holding periods of attention trades of 7

Table 10
Attention trades and individual investor behavior.

		HPR	Leverage	Total volatility	Holding period	Investment
(1)	Open and close attention trades	−4.65	2.61	0.25	0.70	22.88
(2)	Only open attention trades	−14.74	1.52	0.18	23.14	15.51
(3)	Open attention trades	−7.66	2.15	0.22	7.38	19.81
(4)	Close attention trades	38.96	1.95	0.12	19.16	14.19
(5)	Open other stocks attention dates	2.92	1.92	0.11	13.87	13.47
(6)	Close other stocks attention dates	2.32	2.03	0.12	13.56	14.44
(7)	Attention stocks other dates	3.46	2.45	0.12	12.84	17.61
(8)	All stocks all dates	2.82	2.03	0.12	13.64	14.42

This table shows descriptive statistics of individual investor behavior during attention shocks. Variables include holding period returns, leverage, total volatility (leverage multiplied with the stock's volatility), holding period (time difference in days between opening and closing of positions), and investment weight (portfolio share invested into each trade) for each subclass of trades.

days (3) compared to the overall average of approximately 13 days also speak to speculative behavior.

In sum, we provide further evidence in favor of H1 and H2 and conclude that high-attention trades increase investor risk-taking and harm investor welfare. Investors engage in speculative behavior through higher total volatility, investments, and shorter holding periods in assets and suffer financial consequences. However, investors do not increase their risk-taking with borrowed money. Leverage is not higher for attention trades.

6.5. Retail trading and social media sentiment

In H3, we hypothesize that emotional sharing is a predictor of retail trading patterns. In H4, we hypothesize that analyst investment advice cannot predict investor trading patterns. Hence, in this subsection, we investigate the predictive power of multidimensional investor sentiment (emotions) derived from WSB posts on different proxies for retail trading while controlling for buy and sell recommendations of analysts. We aim to investigate whether sentiment as opposed to analyst investment advice has predictive power for retail trading. Specifically, we estimate the following model:

$$y_{i,t} = \alpha + \sum_{j=1}^{J} \beta_j \; sentiment_{ij,t-1} + \gamma \; analysts_{i,t-1} + \sum_{k=1}^{K} \delta_k \; stock \; controls_{ik,t-1} + \sum_{n=1}^{N-1} \zeta_n \; weekday_t + \sum_{p=1}^{P-1} \eta_p \; stock_i + \epsilon_{il,t}, \tag{8}$$

where $y_{i,t}$ is either the buy–sell imbalance of both long and short positions, of long positions, or of short positions for firm i during day t. The variables of interest are Ekman's six emotions, which include *joy*, *surprise*, *anger*, *disgust*, *sadness*, *fear*, and *neutral*, and the analysts buy and sell recommendations, which are all lagged by one day. Stock controls include *BSI*, *event buzz*, *sentiment*, *return*, *market capitalization*, and *price-to-book ratio* for firm i during day $t-1$. We show the results in Table 11.

The results provide support for both Hypotheses 3 and 4. Analyst recommendations have no predictive power for the two variants of retail trading. In contrast, we observe predictive power for user contributions that contain the emotions anger and disgust. This result provides indicative evidence that it is not only attention that drives retail trading patterns, but that the type of attention matters. This evidence is in line with Duz Tan and Tas (2021), who show that strong negative emotions such as anger and disgust are negatively correlated with the absolute buying of stocks.

Note that the multidimensional sentiment variables, which are derived from WSB posts, have predictive power, albeit controlling for overall market sentiment and event buzz. While event buzz positively predicts buy–sell imbalances, the coefficient on overall market sentiment (ESI) is not significantly different from zero. This is suggestive evidence that overall market sentiment and retail investor sentiment differ.

Overall, the results in this section contribute to the growing literature on the impact of emotions on financial decision-making and financial markets (Vamossy, 2021, 2024). Overall, this literature comes to different conclusions as to whether positive or negative emotions have a greater influence. Some studies document significant relationships between positive and negative emotions and investor behavior and market outcomes (Griffith, Najand, & Shen, 2020; Shen, Griffith, Najand, & Sun, 2021; Shen, Najand, Dong, & He, 2017), other contributions in this literature find positive emotions such as trust and optimism to have a larger impact on investors' decisions than negative emotions (Bird, Gallagher, Khan, & Yeung, 2023). Our results align with Nguyen et al. (2020) who show that transient investors decrease their stockholdings in response to increases in negative emotions such as anger, disgust, and fear.

7. Implications and conclusions

This paper analyzes whether WSB attention influences investor risk-taking and welfare and spurs uninformed trading by retail investors. The analyses combine individual investor data from an online brokerage service with retail investor attention that is derived from the subreddit WallstreetBets (WSB). While investors seem to refrain from leveraging their trades on high-attention days, they engage in speculative behavior that increases their overall risk levels. We show that WSB attention decreases holding period returns of investors for short sellers and other investors alike. Finally, we show that investment decisions of meme investors are more likely to be based on emotional responses to angry and disgusted user contributions on social media rather than analyst investment advice.

Our results have important practical implications. We show that meme investors, on average, engage in speculative behavior when trading on attention (which they do) and lose significant amounts of money doing so. Thereby, we provide support for regulators' growing concern that social media and "finfluencers" negatively affect retail investors performance. Future research should investigate the profiteers of meme-stock manias.

Despite the contribution to the literature and practical implications, our study has some caveats. Although the account-level data allow us to study important questions, it is limited to one provider. Although the sample size is more than sufficient to establish statistical power, unique features of the brokerage service might attract specific characteristics in customers or promote certain behaviors that could bias our results. A similar sample self-selection bias pertains to WSB. In particular younger individuals with less investment experience may gather information from social media. As a result, our results may overstate the impact of social media attention on overall retail investor behavior.

Table 11
Retail trading and social media sentiment.

	Dependent variables		
	BSI (1)	BSI long (2)	BSI short (3)
Joy_{t-1}	−0.0521	−0.0005	−0.0003
	(−1.5855)	(−1.5904)	(−0.7571)
$Surprise_{t-1}$	0.0134	0.0001	0.0001
	(0.6676)	(0.6508)	(0.4575)
$Anger_{t-1}$	−0.0137***	−0.0001***	−0.0001***
	(−5.3556)	(−4.9137)	(−3.0594)
$Disgust_{t-1}$	−0.1725**	−0.0019**	0.0012
	(−2.2361)	(−2.5767)	(0.7740)
$Sadness_{t-1}$	−0.0117	−0.0001	−0.0001
	(−0.2951)	(−0.3109)	(−0.1804)
$Fear_{t-1}$	−0.0378	−0.0004	−0.0007
	(−0.3560)	(−0.3192)	(−0.7917)
$Neutral_{t-1}$	0.0203*	0.0002*	0.0001
	(1.7920)	(1.7916)	(1.1105)
$Analysts_{t-1}$	−0.9673	−0.0068	−0.0183
	(−1.4030)	(−0.9385)	(−1.5266)
BSI_{t-1}	0.1537***		
	(8.4599)		
BSI $long_{t-1}$		0.1638***	
		(9.1512)	
BSI $short_{t-1}$			−0.0906***
			(−14.3019)
Event $buzz_{t-1}$	0.3430***	0.0024**	0.0065***
	(3.0463)	(2.0232)	(3.6005)
ESI_{t-1}	−0.2992	−0.0038	−0.0138
	(−0.4283)	(−0.5243)	(−1.4834)
$Return_{t-1}$	−12.2336	−0.1674**	0.0741
	(−1.5511)	(−2.0484)	(1.1182)
Market cap_{t-1}	0.000003	0.000000	−0.000000
	(1.1185)	(0.8155)	(−1.4132)
Price to $book_{t-1}$	−0.0013	−0.00001	0.0001***
	(−1.3342)	(−1.1534)	(3.1725)
Stock effects	Yes	Yes	Yes
Weekday effects	Yes	Yes	Yes
Stocks	259	259	259
Observations	48,480	48,480	48,480
Adj. R^2	0.0503	0.0546	0.0079

*p<0.1; **p<0.05; ***p<0.01.
This table shows the results of multivariate panel regressions. The dependent variables are daily buy–sell imbalances, which are computed based on both long and short positions (Column 1), only long positions (Column 2) and only short positions (Column 3). Regressors include multidimensional retail sentiment (joy, surprise, anger, disgust, sadness, fear and neutral), analyst buy and sell recommendations, event buzz (number of standard deviations the daily event volume of a company is above or below its mean), sentiment (average sentiment of events per day), stock return, market capitalization and price-to-book ratio. We also include individual stock and day-of-the-week effects. Regression coefficients are presented with *t*-statistics in parentheses. Standard errors are double-clustered (time and firm).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Table A.1.

Data availability

The data that has been used is confidential.

Table A.1
WSB attention and other attention measures.

Panel A: Pearson correlation coefficients		
	Social attention signal	WSB
ASVI	0.16	0.18
Social signal	–	0.21
Panel B: Regression analysis of WSB attention		
	WSB	
(Intercept)	−6.378	
	(−1.263)	

(*continued on next page*)

Table A.1 (*continued*).

ASVI	9.640
	(2.495)
Social attention signal	54.596
	(4.278)
Obs.	28,272
Adj. R^2	0.141

Panel A shows Pearson correlation coefficients for Google Search Volume (*ASVI*, Da et al., 2011), the social attention signal from Cookson et al. (2024) and ticker/company name mentions on WSB (WSB). Panel B reports the results from a regression analysis on WSB. Data is from January 1, 2020, to December 31, 2021. *t*-statistics are in parentheses.

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